Tuesday, March 4, 2025

Section 5.1

LEC22 - Sequences

Warm-Up Problem

1. Which of the options below could have the graph on the right? d)
2. Is the sequence in the graph convergent or divergent? What is the limit? The sequence appears to be convergent to $1/2$

   a) $a_n = \frac{1}{n}$ b) $a_n = \frac{(-1)^n}{n}$ c) $a_n = 1 - \left(\frac{5}{6}\right)^n$ d) $\frac{1}{2} + \left(\frac{5}{6}\right)^n$

Example

Are the following statements true or false?

1. $\{a_n\}_{n=1}^{\infty}$ converges iff $\{a_n\}_{n=5}^{\infty}$ converges. True
2. Suppose $b_n = c_n$ for $n > 5$ (but possibly different for $n \le 5$). Then $\{b_n\}_{n=1}^{\infty}$ converges iff $\{c_n\}_{n=1}^{\infty}$ converges. False

Example

Compute the following three limits.

1. $\lim_{n\to\infty} \sqrt{\frac{3n^3+2n}{4n^3-10}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$
2. $\lim_{n\to\infty} \frac{e^{2n}}{1-n} \overset{H}{\Rightarrow} \frac{2e^{2n}}{-1} = -\infty$
3. $\lim_{n\to\infty} \frac{\pi^{n+1}(\sqrt{2})^{2n}}{6^{n-1}} = \frac{\pi^2 \cdot \pi^{n-1} \cdot 2 \cdot 2^{n-1}}{6^{n-1}} = \overbrace{2\pi^2 \underbrace{\left(\frac{2\pi}{6}\right)}_{r}^{n-1}}^{\text{Geometric Sequence}} = \infty$ since $r > 1$

Factorial Notation

$5! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$

$n! = 0 \cdot 1 \cdot 2 \cdot \dots \cdot n$

$1! = 1$

$0! = 1$

Example

Find the limit of the sequence $\{a_n\}$ where $a_n = \frac{n!}{(n+2)!}$

$\Rightarrow \frac{n!}{(n+2)!} = \frac{\cancel{n!}}{\cancel{n!} \cdot (n+1) \cdot (n+2)} = \frac{1}{(n+1)(n+2)} \Rightarrow \lim_{n\to\infty} \left[\frac{1}{(n+1)(n+2)}\right] = 0$

Example

Find the limit of a sequence $\{a_n\}$, where $a_n = \frac{\sin(n)}{n^2}$

$\Rightarrow$ We can use the squeeze theorem

$\Rightarrow -1 \le \sin(n) \le 1$

$\Rightarrow -\frac{1}{n^2} \le \frac{\sin(n)}{n^2} \le \frac{1}{n^2}$

$\Rightarrow \lim_{n\to\infty}\left[-\frac{1}{n^2}\right] \le \lim_{n\to\infty}\left[\frac{\sin(n)}{n^2}\right] \le \lim_{n\to\infty}\left[\frac{1}{n^2}\right]$

$\Rightarrow 0 \le \lim_{n\to\infty}\left[\frac{\sin(n)}{n^2}\right] \le 0$

$\Rightarrow \lim_{n\to\infty}\left[\frac{\sin(n)}{n^2}\right] = 0$